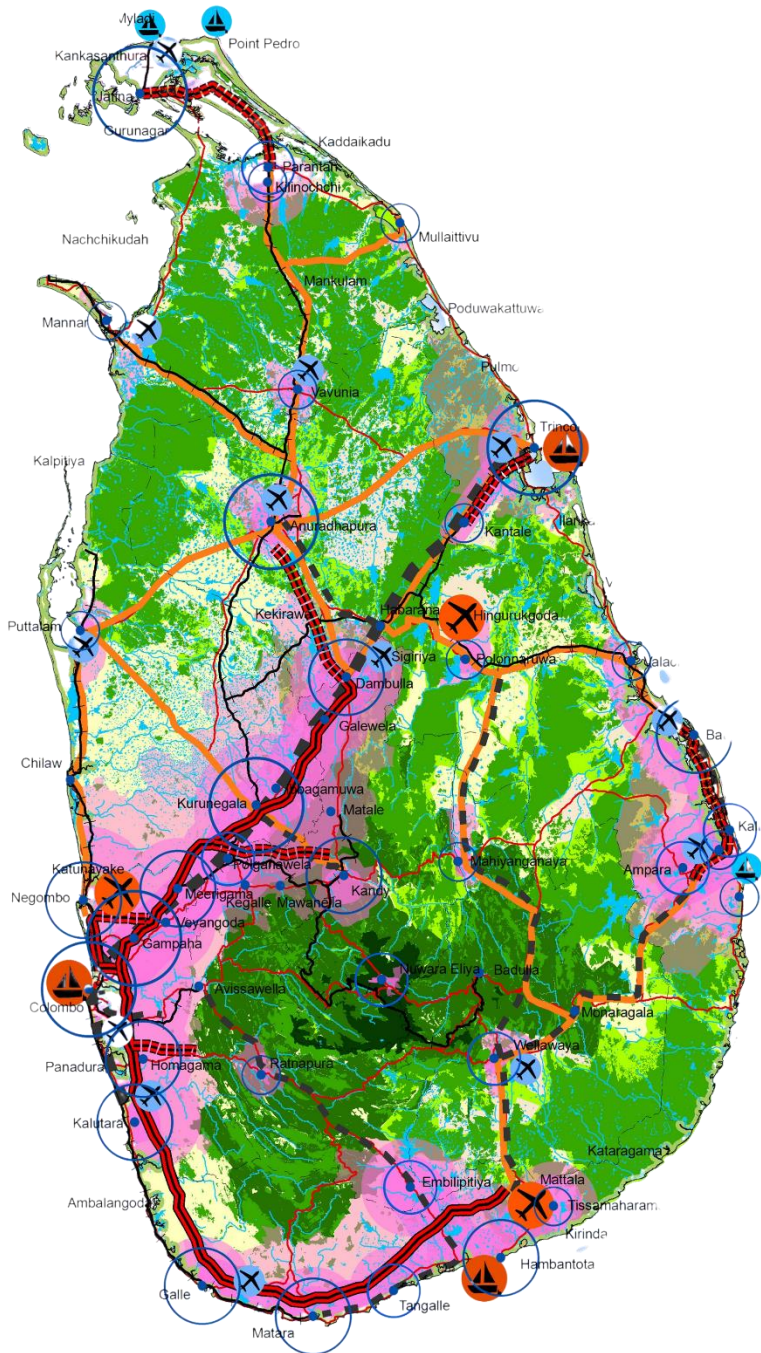


Policy Brief -

Bridging Gaps in the National Physical Planning Policy: Strategic Recommendations to Address the Human-Elephant Conflict.



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1. Introduction

I. Overview of the Issue.

The Human Elephant Conflict (HEC) is a significant issue in Sri Lanka, arising from overlapping human settlements, agricultural lands, and elephant habitats, leading to frequent encounters and conflicts.

II. Purpose.

This report aims to analyze land use conflicts contributing to HEC using the Land Use Conflict Identification Strategy (LUCIS) model, assess gaps in the National Physical Planning Policy (NPPP), and recommend improvements for sustainable development.

III. Policies under Review.

- National Physical Planning Policy (NPPP)
- Proposed conservation zones under the National Physical Plan (NPP)

2. Background.

HEC persists due to the lack of alignment between conservation and development strategies in Sri Lanka. This analysis explores the limitations of the NPPP, particularly in sensitivity mapping and urban planning frameworks, in addressing this issue.

I. Thesis/Research Question.

How can Sri Lanka's planning policies be adapted to better address the land use conflicts contributing to the Human Elephant Conflict?

II. Policy Context and Background.

➤ Historical Background and Policy Development.

Sri Lanka's National Physical Planning Policy (NPPP) has evolved to guide land use and development but lacks specific measures for HEC. Historical interactions and agricultural expansion have exacerbated these conflicts.

➤ **Context of Implementation.**

Implemented to guide sustainable land use, the NPPP currently fails to account for HEC hotspots within its sensitivity mapping, potentially leading to inadvertent developments in highrisk zones.

III. Key Stakeholders

- National Physical Planning Department officials
- Wildlife conservation groups
- Regional planning authorities
- Urban Development Authority (UDA) and Divisional Secretariat Division (DSD)

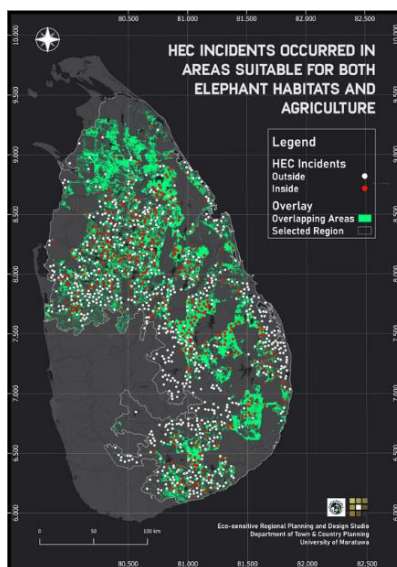


Figure:01

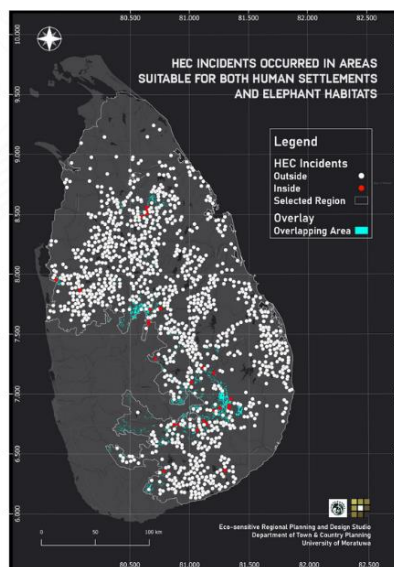


Figure:02

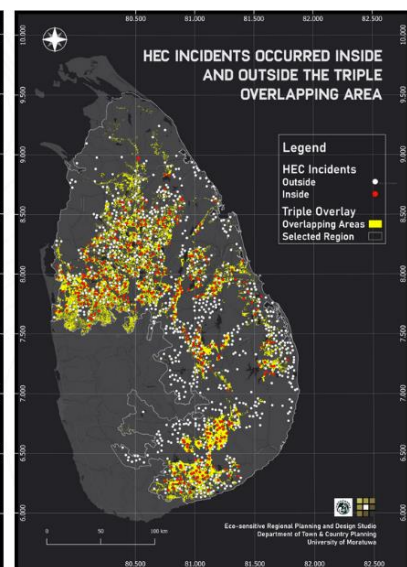


Figure:03

3. Policy Analysis Framework

I. Framework/Methodology

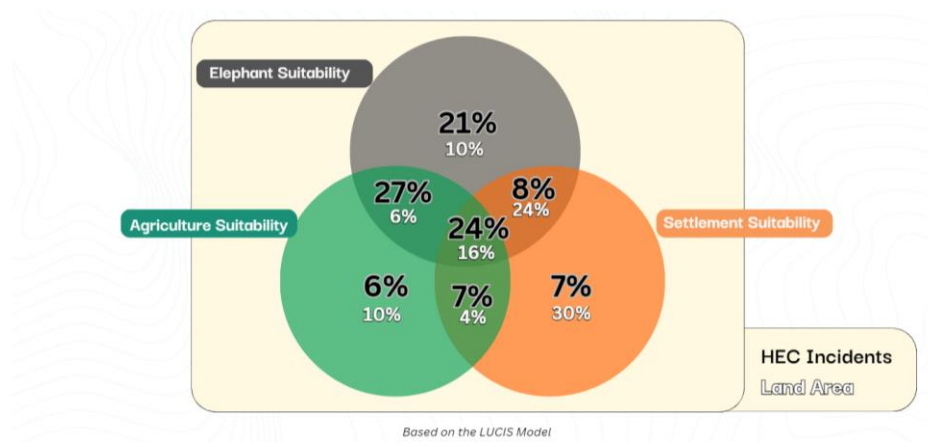
The LUCIS model is used to assess land suitability overlaps for settlements, agriculture, and elephant habitats, identifying conflict zones.

II. Key Concepts

- **HEC Density:** HEC Density is a metric that quantifies the frequency of humanelephant interactions within specific areas, often indicating highrisk zones for conflict.
- **Sensitivity Mapping:** Sensitivity Mapping is a spatial tool used to identify areas vulnerable to environmental impacts and conflicts. It integrates ecological factors to guide landuse planning decisions, helping to protect critical habitats and manage HEC hotspots
- **Urban Hierarchy and Infrastructure:** Urban Hierarchy classifies urban areas based on factors like livability, development potential, population density, and ecological sensitivity, guiding the intensity and type of development appropriate for each area.

III. Evaluation Criteria

- Alignment with conservation objectives
- Minimization of humanelephant conflict
- Balancing urban development with environmental sensitivity



4. Policy Goals and Objectives

I. Goals of the NPPP

The NPPP aims for sustainable development across Sri Lanka but lacks explicit goals related to HEC mitigation.

II. Evaluation of Goals

The current goals do not adequately address the ecological and social challenges posed by HEC.

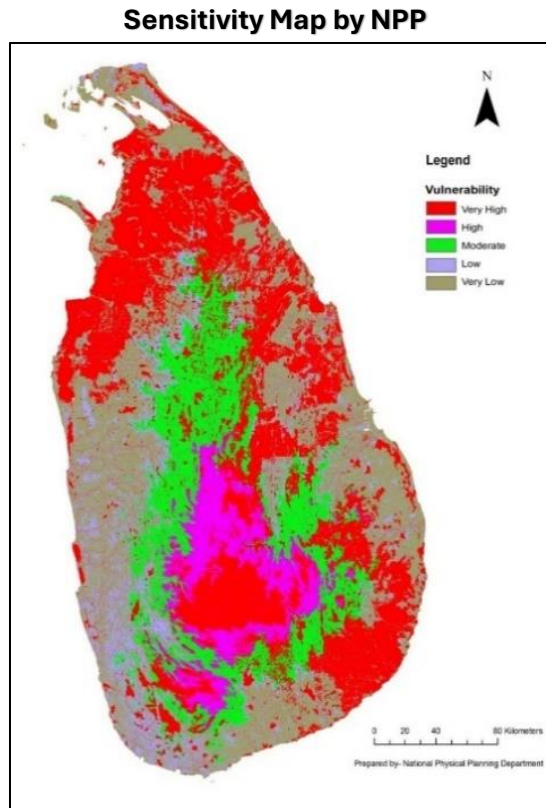
III. Conflicts in Goals

1. Adjustable Sensitivity Map for Development and Conservation

The current sensitivity map within the NPPP does not account for HEC density, resulting in some high HEC areas being inaccurately classified as low sensitivity zones. This misclassification leads to development in areas prone to HEC, intensifying conflict between human settlements and elephant habitats.

Attribute	Criteria for Evaluation	Rank in the order of Relative Importance
Water resources	The level of influence on the catchments of the major rivers, tanks and reservoirs and the quantity of flows.	1
Wildlife and biodiversity	The support provided for the sustenance of wildlife habitats and the biodiversity and the extents of lands.	2
Forest, flora and fauna	The type and the extent of ground cover provided, and the variety added to the overall landscape.	2
Ecosystems	The level of contribution to coastal and riverine ecosystems and the traditional livelihood of communities	3
Archeological areas	The archeological value and the association with the history of the island.	4
Unique landscapes	The level of attraction by the tourists and the contribution to the image.	4
Agricultural uses	The level of productivity of the lands in terms of conventional agricultural uses.	5

Table:01



2. Urban Hierarchy and Regional Imbalances

The current urban hierarchy system within the NPPP applies uniform criteria to rank urban areas, focusing primarily on livability factors. This approach overlooks regional HEC density, leading to potential development imbalances. High ranking urban centers in high HEC density areas are often promoted for urban expansion, which may increase human elephant encounters and degrade sensitive habitats.

Attribute	Criteria for evaluation	Rank in the order of Relative Importance
Availability of developable land	Extents of land free from agricultural uses, conservation, not prone to landslides, less exposed to floods, cyclones and other disaster situations.	1
Availability of safe drinking water and water for other needs	Present and future availability of water in the respective river basins.	2
Proximity to urban centers	Distance to urban centers	3
Access to electricity	Distance to the national grid	4
Impact of major infrastructure developments	Distance from the existing or proposed expressways, highways, water supply schemes, etc.	4
Average daytime temperature	Variation from the mean 2025 Degree Celsius	5
Annual Rainfall	Variation from the mean 2000 3000 mm	6

Table:02

Appendix 5.3.2.f : Proposed Urban Hierarchy - 2050

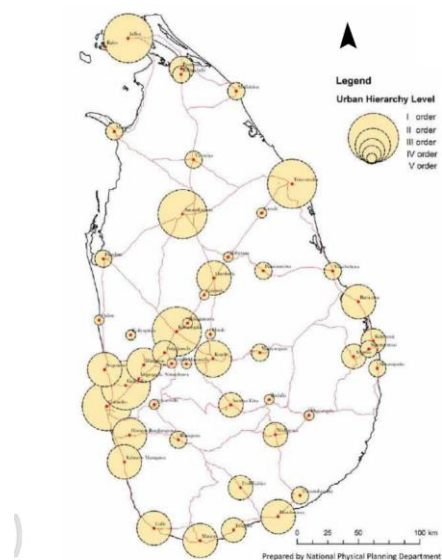


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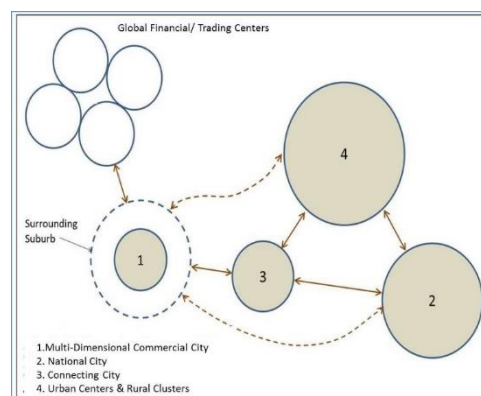


Figure:07

2. Infrastructure Development in Conservation Zones

Infrastructure projects, including proposed electrified railways, road expansions, and airport developments, aim to stimulate economic growth and connectivity. However, some of these projects overlap with designated elephant conservation zones, **directly conflicting with conservation objectives** by intruding upon sensitive habitats.

5. Policy Implementation

I. Current Implementation

The National Physical Planning Policy (NPPP) currently utilizes sensitivity mapping and an urban hierarchy to guide land use and urban growth. However, it lacks integration of Human-Elephant Conflict (HEC) data, limiting its effectiveness in addressing HEC issues within designated areas.

II. Challenges

- Gap Among Authorities Managing Land and Community Relations
 - Key local authorities, such as Divisional Secretariats (DSD) and Urban Development Authorities (UDA), lack sufficient awareness and understanding of NPPP guidelines. This gap weakens the implementation of conservation measures and coordination with local communities.
- Lack of Local Awareness of NPPP Designations
 - Many local communities are not well informed about the conservation and development zones outlined by the NPPP, which impacts adherence to guidelines and increases risks of human wildlife conflict.
- Infrastructure Encroachment into Conservation Zones
 - Development often extends into conservation areas, leading to habitat fragmentation and exacerbating the HEC.

III. Evaluation of Implementation Strategies

Current strategies lack a coordinated approach that aligns conservation and urban development objectives, resulting in escalating HEC issues. Bridging gaps between conservation authorities, urban planners, and local land management bodies is essential to foster an integrated response to HEC within the framework of urban development.

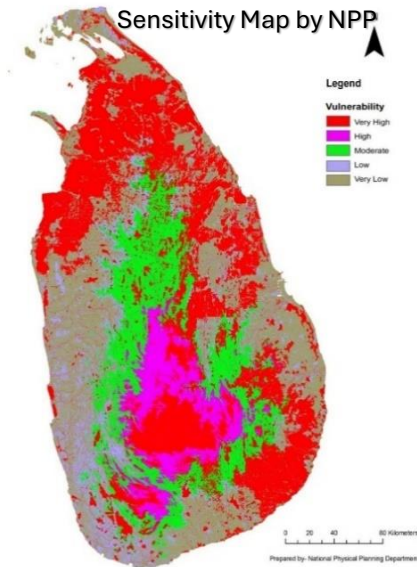


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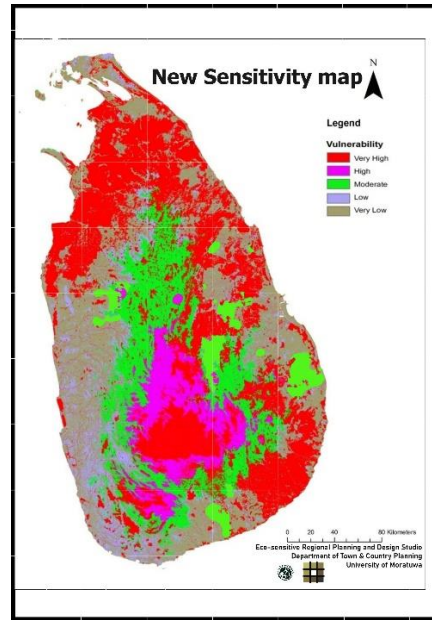


Figure:09

6. Policy Outcomes and Impacts

I. Gaps and Impacts

1. Sensitivity Mapping and Human-Elephant Conflict

The NPPP includes a sensitivity map designed to guide development by indicating areas of environmental vulnerability. However, this map does not consider HEC as a factor, leading to discrepancies in identifying sensitive areas related to human wildlife conflicts. When we overlaid our HEC density map onto the NPPP's sensitivity map, it revealed that several areas classified as low sensitivity zones experience high HEC incidents.

This oversight in the NPPP suggests that the sensitivity map underrepresents the actual risks of HEC, potentially leading to development approvals in regions that are highly vulnerable to human-elephant interactions. The absence of HEC data in sensitivity mapping thus constitutes a gap in the policy, undermining its effectiveness in addressing this critical conflict.

2. Urban Hierarchy Map

The NPPP's urban hierarchy map ranks urban areas based on livability factors alone, without integrating sensitivity or development potential metrics. As a result, some high ranking urban areas, such as Anuradhapura, Ampara, and Hambantota, overlap with high HEC density regions.

Assigning similar hierarchy levels to areas with vastly different population densities and environmental sensitivities may lead to development imbalances, where regions with high conservation needs are exposed to intensified urbanization and infrastructure demands. This lack of integration between livability, sensitivity, and development potential results in an urban hierarchy that may exacerbate HEC in sensitive regions.

3. Infrastructure Projects in Conservation Zones

The NPPP's 2050 infrastructure plan includes several projects that encroach upon elephant conservation zones. Notably, an electrified railway network is proposed from Dambulla to Trincomalee, crossing regions identified as high HEC density zones and environmentally sensitive areas. Similarly, proposed rail networks for agricultural logistics in the southern and eastern regions, and certain airport developments, are situated within or near conservation zones.

These infrastructure plans highlight a significant gap in the NPPP's alignment with conservation priorities. Expanding infrastructure into conservation areas can exacerbate habitat loss, increasing the likelihood of HEC as elephants are forced into human inhabited areas. By not adequately accounting for HEC density in the location of infrastructure projects, the NPPP risks intensifying human wildlife conflicts and undermining long term conservation efforts.

II. Outcomes

The NPPP's gaps in HEC related mapping have led to unintended consequences, with development permitted in conflict prone areas.

III. Achievement of Goals

The policy's failure to address HEC directly limits its success in achieving sustainable land use and biodiversity protection.

IV. Unintended Consequences

Infrastructure projects within conservation areas disrupt elephant habitats, further escalating HEC.

7. Policy Alternatives

- **Alternative 1: Adjusted Sensitivity Mapping**

Updating sensitivity maps to include HEC data could help avoid development in high conflict zones.

- **Alternative 2: Differentiated Urban Hierarchy**

Refining urban hierarchy levels based on ecological sensitivity and projected population density would promote balanced growth.

- **Alternative 3: Elevated Infrastructure Solutions**

Converting railway lines to elevated tracks and exploring drone logistics could reduce land use in conservation zones.

Current Approach

Figure 5.1 : The Proposed Spatial Structure - 2050

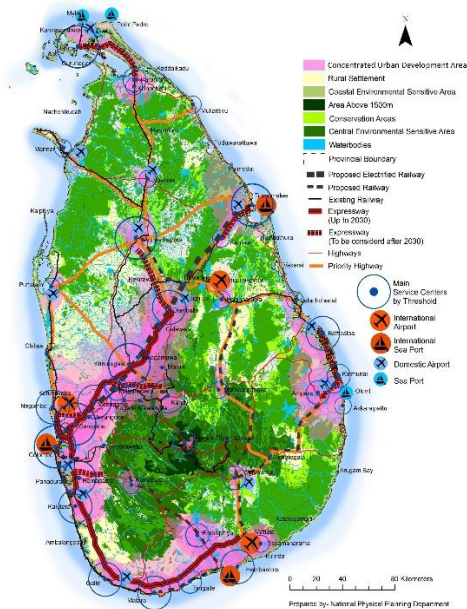


Figure:10

Alternative Approach

Modified National Physical Plan

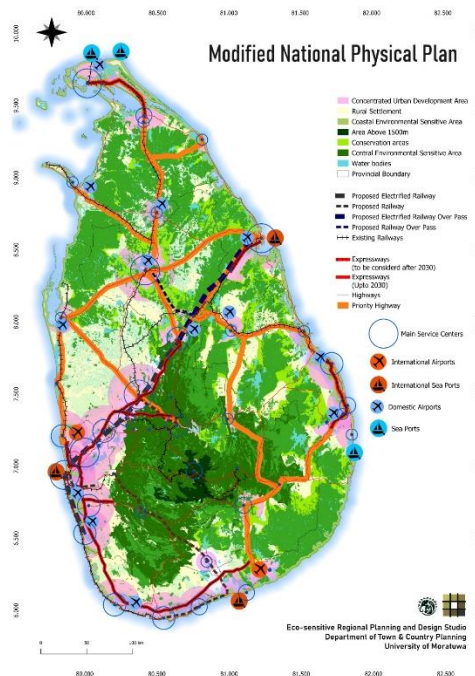


Figure:11

8. Recommendations

I. Enhance Conservation Zones in the NPP

The National Physical Plan (NPP) of Sri Lanka proposes a conservation zone dedicated to elephants, intended to mitigate the escalating human elephant conflict and secure a sustainable habitat for the elephant population into the future. To evaluate the adequacy of this proposed conservation area for supporting the projected elephant population by 2050, we undertook a two part analysis. First, we projected the elephant population for 2050 using a logistic growth model well suited for estimating populations constrained by environmental resources. This approach projected a population of approximately 12,000 elephants by midcentury. Subsequently, we calculated the carrying capacity of the proposed conservation area, considering the varying capacities associated with different forest types within the zone. This detailed assessment revealed a carrying capacity of roughly 16,000 elephants, indicating that the proposed area is more than sufficient to support the anticipated population.

Despite the encouraging data, field visits exposed an emerging threat to the conservation zone's integrity. We observed several ongoing development activities within the conservation area, likely stemming from limited awareness among local authorities, such as the Divisional Secretariat Division (DSD) and the Urban Development Authority (UDA), regarding the NPP's designated land use allocations. Since each forest type within the conservation zone contributes a unique carrying capacity, any reduction in forest cover or changes in forest composition could significantly compromise the area's overall capacity to support the elephant population. If urbanization within the conservation zone remains unchecked, it could erode the intended conservation benefits, reducing the carrying capacity and potentially leading to negative ecological consequences.

In light of these findings, we recommend suspending all urbanization activities within the conservation zone to preserve its ecological role. Raising awareness among local authorities and stakeholders about the NPP's objectives and land use designations is crucial to achieving this goal. Although the NPP is currently gazetted at the national level, extending this gazette to the local level would provide local authorities with a stronger mandate to enforce development restrictions in line with conservation objectives. By implementing these measures, the NPP can help protect the conservation zone's carrying capacity, ensuring it remains a viable habitat for Sri Lanka's elephant population through 2050 and beyond.

II. Updating Sensitivity map.

Updating the NPPP's sensitivity map's criteria by including HEC density data will provide a more accurate reflection of conflict prone areas. By categorizing high HEC regions as high sensitivity zones, the revised sensitivity map can guide development away from these areas, reducing the likelihood of human elephant interactions. This update would also support targeted interventions for conflict mitigation, such as increased fencing or designated conservation corridors.

III. Recommendations for Adjusting the Human Settlement Hierarchy in the National Physical Plan (NPP)

The National Physical Plan (NPP) of Sri Lanka currently employs a standardized approach to the Human Settlement Hierarchy, assigning similar thresholds for population density and development potential across various cities, irrespective of their unique characteristics and challenges. This approach, however, overlooks critical variables such as location sensitivity, Human-Elephant Conflict (HEC) density, and livability factors, leading to inefficiencies in urban planning and settlement sustainability.

To address these gaps, we conducted an in-depth analysis of settlement hierarchies within high HEC density regions, particularly focusing on cities like Hambantota and Galle. While both cities were assigned the same population density thresholds under the current NPP guidelines, our findings revealed stark differences in their respective HEC impacts, urban development needs, and population structures. For instance, Hambantota experiences significantly higher HEC density, which warrants tailored planning that considers both conservation needs and sustainable urban expansion.

Our analysis proposed a recalibration of the settlement hierarchy by introducing differentiated population density thresholds based on a multicriteria approach. This new approach integrates HEC density metrics, ecological sensitivity, and livability considerations to establish more context sensitive guidelines. By applying this refined hierarchy, development plans can better reflect the unique challenges faced by each city, ensuring balanced growth that minimizes conflicts between human settlements and conservation efforts.

The urban hierarchy map should consider both development potential and environmental sensitivity alongside livability factors. A graph-based approach, as illustrated in the diagram above, can help balance these factors by identifying zones with low sensitivity and high development potential for urban expansion, while reserving high sensitivity, low potential areas for conservation. For regions with both high sensitivity and high development potential, urban planning should factor in future population projections to avoid overdevelopment.

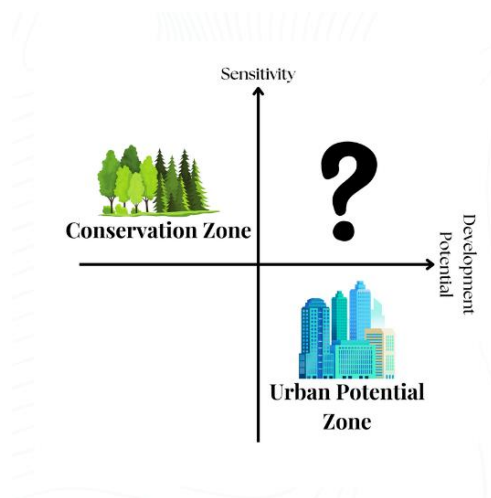


Figure:12 Graph approach

IV. Modify Infrastructure Plans in Conservation Areas

Recommendations for NPPD's Proposed Infrastructure Developments in the Conservation Zone

In reviewing the National Physical Planning Department's (NPPD) infrastructure proposals for 2050, we identified several planned projects within the designated elephant conservation zone. These include the proposed electrified railway network from Dambulla to Trincomalee, a railway network from the southern to eastern region for agricultural logistics, and the promotion of Hingurangoda Airport to international status. Each of these projects could potentially impact the conservation zone's ecological balance and encourage urban development in protected areas, posing challenges to conservation efforts. To address these issues, we propose the following recommendations.

- **Electrified Railway Network from Dambulla to Trincomalee**

The proposed electrified railway line from Dambulla to Trincomalee passes through an environmentally sensitive and high density region within the conservation zone. Ground level infrastructure developments of this scale often draw human activity and settlement, which can drive

urbanization in surrounding areas. To prevent such unwanted development, we recommend converting the railway project into an elevated, overpass railway network. This modification would effectively disconnect the railway from the ground, reducing the likelihood of encroachment and urban sprawl within the conservation zone, thus helping to preserve its ecological integrity.

- **Agricultural Logistics Railway Network in the Southern to Eastern Region**

NPPD has also proposed a railway network from the southern to the eastern and central regions to support agricultural logistics. However, our analysis, which overlays the Human-Elephant Conflict (HEC) density map with the NPP, shows that this railway route would intersect areas with high HEC density. In response, we recommend an alternative logistics solution: establishing a central logistics hub in Mahiyangana to collect, store, and distribute agricultural products. This hub would function as an economic center akin to the Dambulla market, with a focus on minimizing land usage and environmental impact.

Instead of traditional transport methods, we propose using drone technology for agricultural logistics. Unlike conventional air transport, which requires significant land for runways and facilities, drones can operate in compact spaces and provide efficient, flexible transportation for agricultural goods. Current advancements enable drones to carry up to 200 kg, a capacity used in various countries for transporting construction and agricultural materials. By establishing this logistics hub in Mahiyangana region well suited to agriculture—we could fulfill logistical needs while minimizing land and environmental impact, preserving the conservation zone, and avoiding the urbanization risks associated with traditional railway infrastructure.

- **Hingurangoda Airport Development**

NPPD's plan to upgrade Hingurangoda Airport from domestic to international status raises concerns due to its location within the conservation zone. Expanding the airport's status would likely attract more tourists and stimulate development in the surrounding area, promoting urbanization within this ecologically sensitive region. Therefore, we recommend maintaining Hingurangoda as a domestic airport to prevent unnecessary expansion and associated land use that could compromise the conservation zone. Limiting the airport's role to domestic flights would reduce the risk of encroachment and development pressures on the surrounding natural landscape.

- **Oluvil Harbor as logistic transport center.**

To balance development with conservation, we recommend the strategic promotion of Oluvil Harbor as a logistic transport center.

By developing Oluvil Harbor into a primary node for sea mode transportation, it can serve as an efficient logistics hub connecting the eastern region to the Southern Hambantota Logistic Center. This initiative would reduce the dependence on land transport routes that traverse conservation zones, thus mitigating potential conflicts between human activities and wildlife habitats.

Enhancing sea based transportation will not only boost economic efficiency but also provide sustainable support for regional development while protecting conservation zones. This strategic shift would facilitate smoother, ecofriendly logistics and reduce the strain on infrastructure corridors that intersect with sensitive areas, preserving the carrying capacity and ecological integrity of the conservation zones outlined in the NPP.

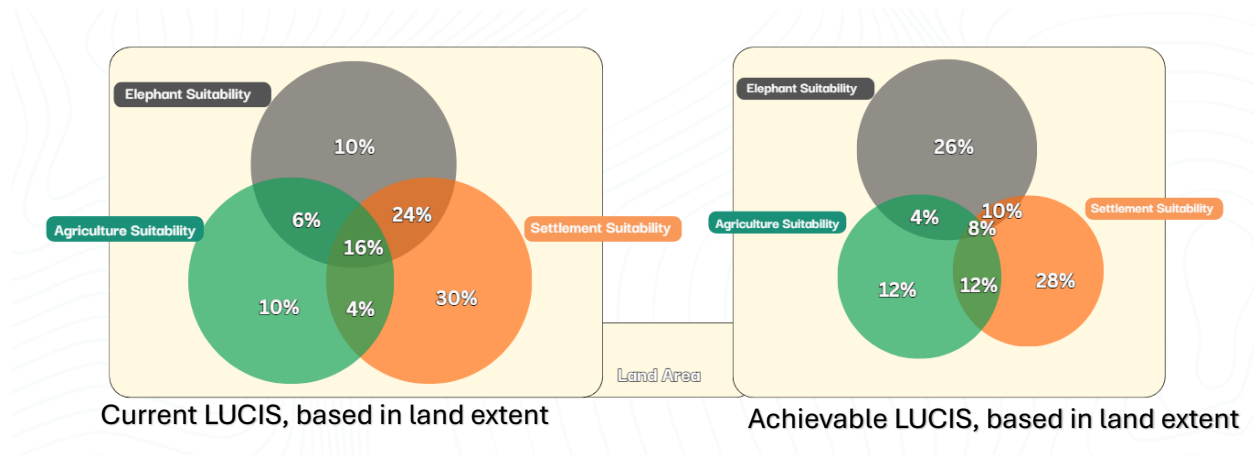


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9. Conclusion

I. Summary

This policy brief emphasizes the need to incorporate HumanElephant Conflict (HEC) data into Sri Lanka's land use planning. By addressing key gaps in the National Physical Planning Policy (NPPP), the country can reduce landuse conflicts, enhance conservation efforts, and support sustainable development.

II. Restatement of Thesis

Aligning the NPPP with ecological realities is crucial to minimizing HEC and fostering balanced growth.

III. Implications

Betterinformed planning can help prevent HEC, protect biodiversity, and ensure longterm urban sustainability for Sri Lanka.

IV. Limitations and Future Research

Further research into the feasibility of drone logistics within Sri Lanka's landscape could offer innovative, ecofriendly solutions to conservation and logistics challenges, enhancing the country's planning and conservation efforts.

10. References

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